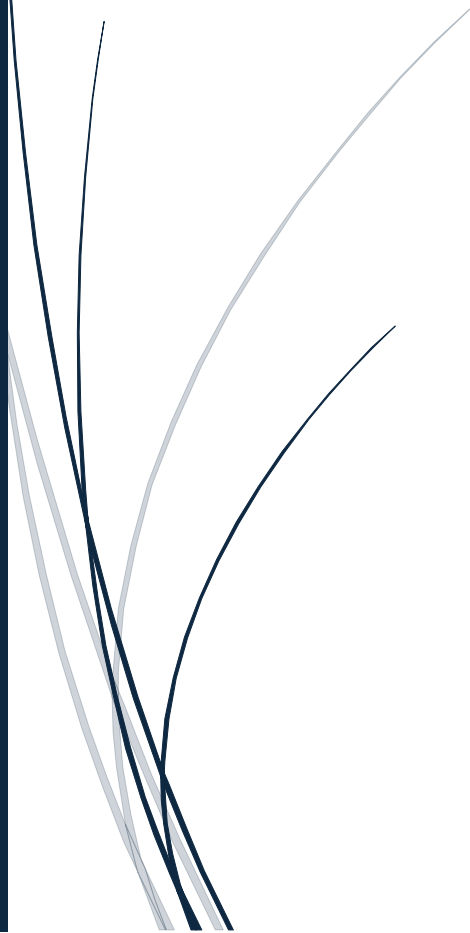




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Drop-off & DNA Rate Analysis

Dataset:- IAPT Waiting Times — NHS



PREPARED BY:- MUKESH RAJ
DATA ANALYTICS INTERN

Introduction

This analysis examines treatment retention, disengagement, and estimated DNA behaviour within NHS Talking Therapies (IAPT) services using the 2024–25 psychological therapies dataset.

The project aimed to:

- estimate DNA rates across NHS commissioning regions,
- analyse treatment drop-off patterns,
- evaluate patient retention through treatment stages,
- and visualise treatment attrition using funnel analysis.

Objectives

The primary objectives of this analysis were:

1. Compute estimated DNA rates by region.
2. Measure treatment drop-off between service access and completed treatment.
3. Analyse retention through treatment stages.
4. Create funnel charts representing patient flow.
5. Export analytical outputs as CSV reports and charts.

Dataset Overview

The merged NHS Talking Therapies dataset contained organisational-level treatment activity and waiting-time indicators.

Key dataset characteristics

Metric	Value
Dataset source	NHS Talking Therapies / IAPT
Analysis period	2024–25
Rows analysed	~92,000
Regions included	NHS Commissioning Regions
Missing-value handling	Applied
Invalid region rows removed	Yes

Data Cleaning and Preparation

The following preprocessing steps were completed before analysis:

- Removed invalid values such as:
 - UNKNOWN
 - All
 - NULL
 - *
 - blank cells
- Converted count variables to numeric format.
- Standardised column names.
- Filtered only NHS Commissioning Regions.
- Removed rows with unusable treatment-count information.

Methodology

Estimated DNA Rate

The dataset did not contain a direct NHS DNA variable.

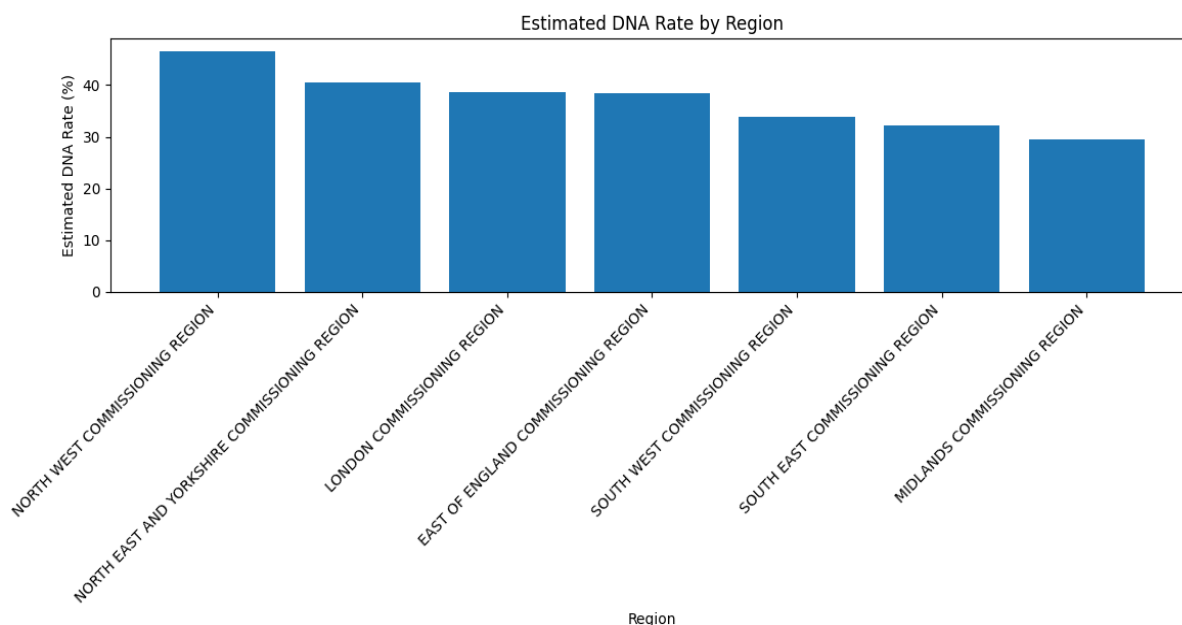
Therefore, an estimated DNA rate was calculated using treatment disengagement between accessed services and completed treatment counts.

The estimated DNA formula used was:

$$\text{Estimated DNA Rate} = \frac{\text{Accessing Services} - \text{Finished Treatment}}{\text{Accessing Services}} \times 100$$

This approach acts as a proxy for patient disengagement and probable non-attendance behaviour.

	Total_Finished_Treatment	Estimated_DNA_Count	Estimated_DNA_Rate
4	2263070.0	1978240.0	46.642193
3	2771295.0	1892670.0	40.580708
1	3110980.0	1954970.0	38.590393
0	2321750.0	1452220.0	38.479903
6	1635560.0	839970.0	33.930916
5	2847020.0	1352220.0	32.201541
2	3625915.0	1509115.0	29.388631

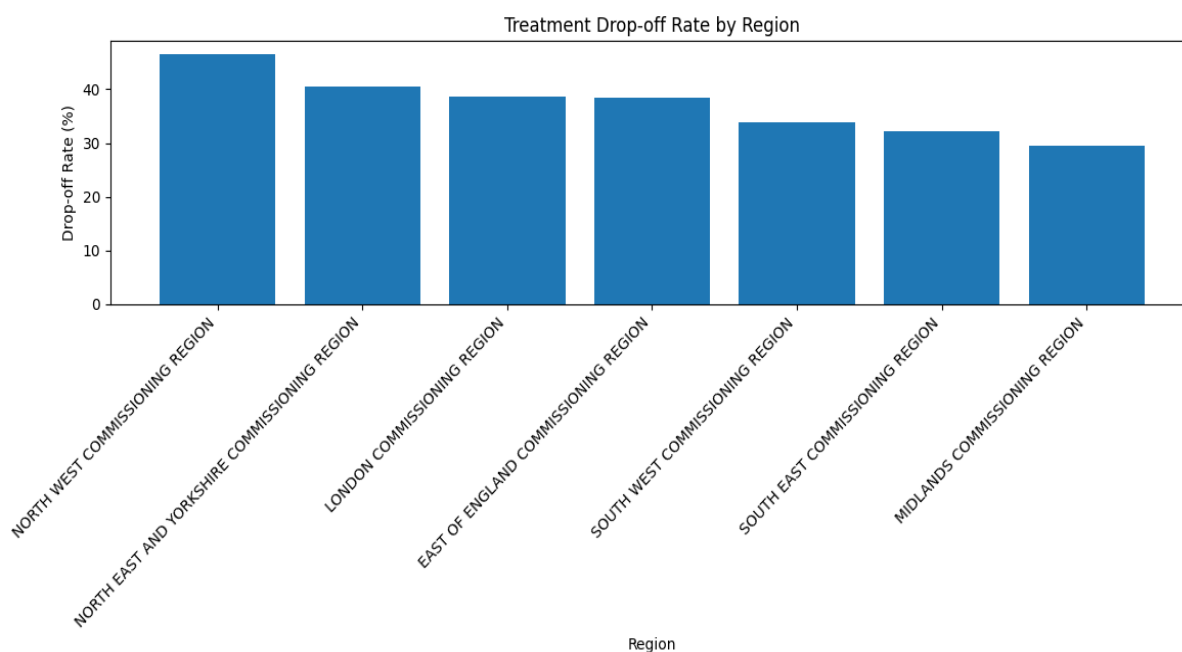


Treatment Drop-off Rate

Treatment drop-off was calculated using:

$$\text{Drop-off Rate} = \frac{\text{Accessing Services} - \text{Finished Treatment}}{\text{Accessing Services}} \times 100$$

This measures the proportion of patients who accessed services but did not complete treatment.



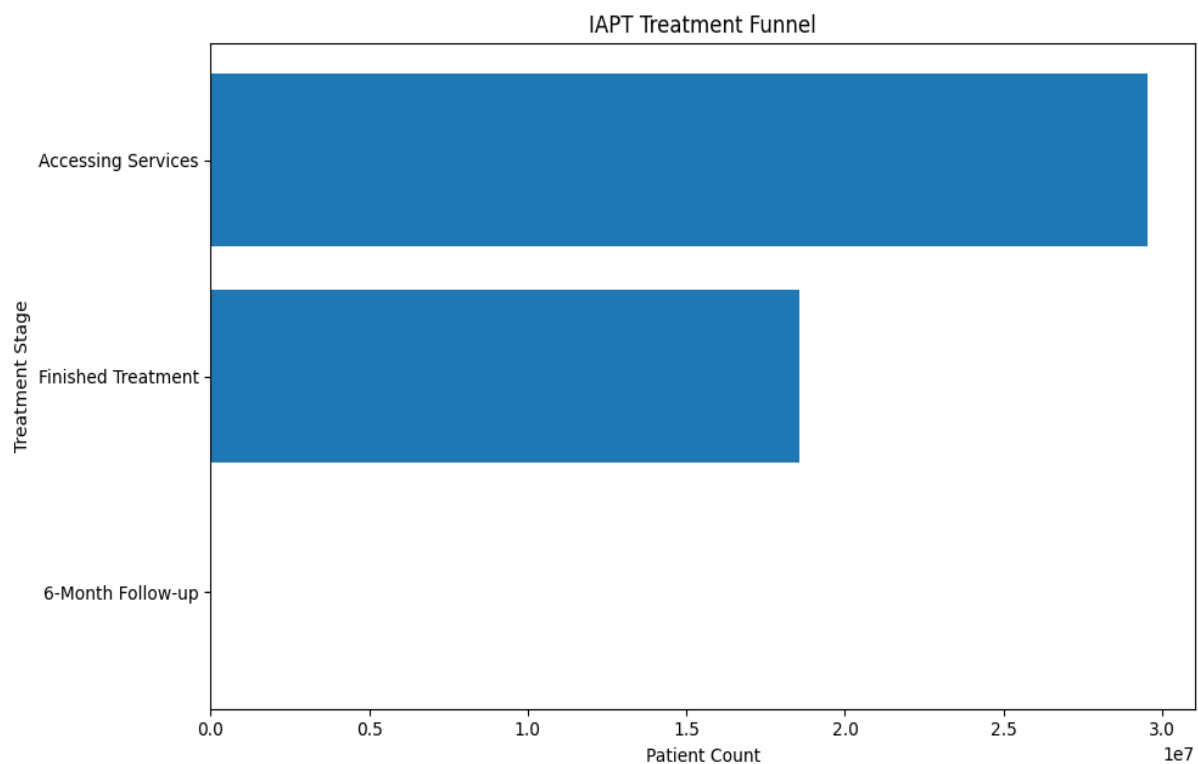
Funnel Analysis

A treatment funnel was constructed using:

1. Accessing Services
2. Finished Course Treatment
3. 6-Month Follow-up

This enabled visualisation of patient attrition throughout the treatment pathway.

	Stage	Count
0	Accessing Services	29554995.0
1	Finished Treatment	18575590.0
2	6-Month Follow-up	5275.0



Regional Estimated DNA Findings

The analysis identified measurable regional variation in estimated DNA behaviour.

Key observations

- Some regions showed substantially higher treatment disengagement rates.
- Regions with lower treatment completion demonstrated higher estimated DNA percentages.
- Regional variability suggests differences in:
 - service accessibility,
 - patient engagement,
 - treatment continuity,
 - or operational performance.

Treatment Drop-off Findings

The drop-off analysis revealed significant reduction between:

- patients accessing services
- and patients completing treatment.

Interpretation

This indicates that a notable proportion of patients discontinue treatment before completion.

Potential contributing factors may include:

- waiting times,
- service accessibility,
- treatment suitability,
- patient motivation,
- scheduling issues,
- or socioeconomic barriers.

Funnel Analysis Findings

The funnel chart demonstrated progressive reduction in patient counts across treatment stages.

Funnel progression

Accessing Services



Finished Treatment



6-Month Follow-up

This indicates that treatment retention may be a critical operational challenge within the therapy pathway.

Visual Outputs Generated

The notebook produced the following outputs:

Output	Description
estimated_dna_rate_by_region.csv	Regional DNA summary
estimated_dna_rate_chart.png	DNA rate chart
dropoff_rate_by_region.csv	Regional drop-off summary
dropoff_rate_chart.png	Drop-off chart
treatment_funnel_summary.csv	Funnel-stage summary
treatment_funnel_chart.png	Funnel visualisation

Limitations

This analysis has several important limitations:

1. Direct NHS DNA variables were not available in the dataset.
2. DNA rates were estimated using treatment disengagement behaviour.
3. Session-level attendance data was limited.
4. Disorder-specific attendance patterns could not be directly evaluated.

Therefore, the estimated DNA analysis should be interpreted as an exploratory operational indicator rather than an official NHS attendance measure.

Conclusion

This analysis identified meaningful treatment attrition patterns within NHS Talking Therapies services.

Key findings include:

- substantial patient drop-off between service access and treatment completion,
- measurable regional variation in estimated DNA behaviour,
- and progressive reduction in patient retention across treatment stages.

The treatment funnel analysis highlighted treatment completion as a major attrition point within the NHS therapy pathway.

Overall, the analysis demonstrates how operational analytics can support:

- service planning,
- patient engagement strategies,
- and NHS performance monitoring.